**Jason Tyler Carlson**
Independent Researcher
New York, NY
www.jasoncarlson.org
917-439-0051
jcarlson212@gmail.com
github.com/jcarlson212

## Education

### Arizona State University - Master of Arts, June 2021

- Major: Mathematics

### Arizona State University - Bachelor of Science, June 2020

- Major: Mathematics
- Second Major: Computer Science
- Outstanding Junior Award: Math Department

## Publications

### Conference Papers

- Carlson, J. T., Bernard, L., Carlson, B. H. (2026). *Beyond Imitation: Preference-Optimized Policies for Realistic Grandmaster Chess*. Proceedings of the IEEE Conference on Games (CoG 2026). Accepted for Oral Presentation. Link.
- Carlson, J. T. and Carlson, B. H. (2025). *The Use of AutoML for Predicting Intracranial Aneurysm Rupture*. Congress of Neurological Surgeons (CNS 2025). Poster. Link.

### Industry Publications

- Moura, J., Carlson, J. T., Singh, J., Xi, S., Koparkar, S., Sun, H., Zhang, W., & Zhang, Z. (2022). *How Amazon Search Reduced ML Inference Costs by 85% with AWS Inferentia. AWS Machine Learning Blog*. https://aws.amazon.com/blogs/machine-learning/how-amazon-search-reduced-ml-inference-costs-by-85-with-aws-inferentia/

### Under Review

- Carlson, J. T. (2026). *TopoExplore: Homology as an Exploration Signal*. Under review at ICLR 2027. Preprint: https://arxiv.org/abs/2607.09971.
- Carlson, J. T. (2026). *Matilda: Engine-Agnostic Search with Human Policy Guidance*. Under review at AAAI 2027. Preprint: https://arxiv.org/abs/2606.25176.

## Fellowships and Awards

- Outstanding Junior Award, Department of Mathematics, Arizona State University, 2019.
- Dean's Scholarship, Arizona State University.

## Professional Experience

**New York University**
*Affiliate Researcher*
Oct 2026–Present

- Working on exploration algorithms for robotic manipulation in the Riviere Robot Lab.

**Independent Researcher**
May 2026–Oct 2026

- Developed TopoExplore, an RL exploration method that uses homology as an exploration signal (under review at ICLR 2027).
- Modeled human move prediction in chess: first-author paper accepted for oral presentation at IEEE CoG 2026, second paper (Matilda) under review at AAAI 2027.
- Built and maintained TopoGym, an open-source benchmark suite for benchmarking exploration on environments varied by topology (used in an RL course at Tufts).

**Amazon**
*Software Engineer II*
May 2025–April 2026

- Led development of machine learning systems for credit extension and risk modeling.
- Built large-scale inference services and ML infrastructure supporting production models.
- Worked on reinforcement learning, foundation model fine-tuning, and distributed ML systems.

**AppLovin**
*Senior Software Engineer*
October 2024–March 2025

- Developed large-scale identity graph infrastructure and distributed data systems.
- Reduced infrastructure costs and improved latency for production ML pipelines.

**Amazon**
*Software Engineer II*
June 2022–September 2024

- Technical lead of a team of six engineers.
- Built a system that groups similar items in the catalog at the Petabyte scale, mentored new engineers, and designed onboarding projects.

**Amazon**
*Software Engineer I*
June 2021–May 2022

- Built production ML data pipelines and large-scale embedding infrastructure.

## Selected Conference and Invited Presentations

- "Beyond Imitation: Preference-Optimized Policies for Realistic Grandmaster Chess," IEEE Conference on Games (CoG), 2026. *(Oral Presentation)*.
- "The Use of AutoML for Predicting Intracranial Aneurysm Rupture," Congress of Neurological Surgeons (CNS), 2025. *(Poster)*.

## Technical Skills

Languages: Python, C++, Java, SQL
ML: PyTorch, JAX, CUDA, distributed training, RLHF/DPO-style preference optimization
RL & Simulation: Gymnasium, MiniGrid, habitat-sim
Topology & Math: GUDHI, persistent homology, algebraic topology, convex optimization

Data & Infrastructure: Apache Spark, Ray, AWS (EC2/ECS/SageMaker/Inferentia), Google Cloud

**Selected Coursework**

Real Analysis I (A), Real Analysis II (A+), Topology I (A+), Topology II (A), Intro to Artificial Intelligence (A), Data Structures & Algorithms (A+), Functional Analysis II (A), Abstract Algebra II (A).

**Research Grants**

- None to date.

**Open Source**

- *GarryChess*–a project for making chess engines play chess more realistically at the high Elo (Grandmaster) band. Project Website. GitHub Organization.
- *TopoGym*–gym environments for reinforcement learning that help train / benchmark topological navigation skills. https://github.com/jcarlson212/TopoGym